EduPro Learning Analytics

Student Segmentation and Personalized Course Recommendation System

Abstract

Online learning platforms serve learners with vastly different goals, backgrounds, and behavioral patterns. Applying a uniform course recommendation strategy across such a heterogeneous population leads to low engagement, poor course completion, and diminished long-term platform loyalty. This paper presents the design, methodology, and empirical results of a Personalized Course Recommendation System for EduPro — a data-driven engine that combines unsupervised learner segmentation with a hybrid recommendation framework. Using K-Means clustering (validated with Hierarchical Clustering) applied to engineered behavioral features — including total courses enrolled, average spending, categorical diversity score, and learning depth index — we identify three distinct learner segments across a cohort of approximately 2,975 users. These segments, termed Engaged Explorers (Cluster 1, n≈457), Moderate Spenders (Cluster 0, n≈1,062), and Casual Learners (Cluster 2, n≈1,468), exhibit clear behavioral separation and call for markedly different recommendation strategies. The system is exposed through an interactive Streamlit web application featuring a learner profile explorer, cluster visualization dashboard, segment comparison panels, and AI-generated actionable insights. Experimental results confirm strong cluster separation (silhouette score ≥ 0.40) and demonstrate that segment-aware recommendations surface meaningfully more relevant content than generic popularity-based baselines

Keywords: *e-learning personalization, learner segmentation, K-Means clustering, course recommendation, behavioral analytics, Streamlit dashboard, EduPro*

1 Introduction

The explosive growth of online education has transformed how learners acquire knowledge and skills. Platforms such as EduPro now host thousands of courses spanning technology, business, creative design, data science, and beyond, enrolling learners from vastly different professional backgrounds, motivation levels, and financial circumstances. Despite this diversity, most platforms continue to rely on one-size-fits-all recommendation engines driven by aggregate popularity metrics — a strategy that fundamentally fails to account for individual learner heterogeneity.

The consequences are well-documented: low recommendation acceptance rates, high course dropout, and premature platform churn. A learner exploring beginner programming courses has little in common with a professional seeking advanced machine learning certifications, yet both may receive identical recommendation sets if the underlying engine lacks a model of learner behavior.

This paper addresses this gap by presenting the design, implementation, and evaluation of a Personalized Course Recommendation System for EduPro. The system applies K-Means clustering to a rich set of engineered behavioral features, identifying three distinct learner segments that serve as the foundation for targeted, segment-aware course recommendations. An interactive Streamlit web application exposes these capabilities to educators and platform administrators through an intuitive, filter-driven interface.

1.1 Research Objectives

- **RO1:** Identify behaviorally homogeneous learner segments using unsupervised clustering
- **RO2:** Engineer a rich feature set capturing engagement, preference, and spending behavior
- **RO3:** Build a hybrid recommendation engine that combines content-based and cluster-collaborative filtering
- **RO4:** Validate cluster quality and recommendation precision through rigorous evaluation metrics
- **RO5:** Deliver system outputs through an interactive, stakeholder-facing Streamlit dashboard

1.2 Paper Organization

Section 2 situates this work within the existing literature on learner modeling and e-learning recommendation. Section 3 describes the dataset and problem context. Section 4 details the feature engineering pipeline. Section 5 presents the clustering methodology. Section 6 describes the Streamlit dashboard and its components, illustrated with screenshots from the deployed application. Section 7 reports experimental results. Section 8 discusses implications and limitations. Section 9 concludes the paper.

2. Background, Context, and Problem Statement

2.1 The EduPro Platform and Learner Diversity

EduPro's course catalog spans at least five primary categories — Design, Machine Learning, Business, Web Development, and Data Science — at three difficulty levels (Beginner, Intermediate, Advanced). The platform's learner base is highly heterogeneous:

- Domain explorers who sample beginner courses across multiple subject areas
- Deep specialists who concentrate enrollments within a single high-value category
- Career-oriented certifiers targeting industry-recognized credentials
- Casual browsers who have not yet found their learning niche

2.2 Problem Statement

EduPro currently applies one-size-fits-all course recommendations across all users, with no structured learner segmentation framework. As a result, learners struggle to discover relevant content, and significant engagement and retention opportunities are lost.

The deficiencies of the current state are threefold. First, recommendations are driven by aggregate popularity rather than individual relevance, producing low acceptance rates. Second, the absence of a behavioral taxonomy prevents targeted marketing, pricing, and content acquisition strategies. Third, learner engagement data — a rich signal of intent — is not systematically leveraged for platform improvement.

2.3 Related Work

Learner modeling has a rich history in intelligent tutoring systems (ITS), beginning with Bayesian Knowledge Tracing (Corbett & Anderson, 1994). More recently, behavioral clustering has been applied to MOOCs to predict dropout (Whitehill et al., 2017) and personalize content delivery (Thai-Nghe et al., 2010). Collaborative filtering — pioneered by the Netflix Prize (Koren et al., 2009) — has been extensively adapted for course recommendation. Hybrid approaches combining content-based and collaborative signals (Burke, 2002) consistently outperform either paradigm alone. Our work extends this tradition by grounding recommendations in learner segments derived from behavioral clustering, rather than relying on explicit ratings or user-item co-occurrence alone.

3. Dataset Description

The EduPro dataset consists of three relational sheets that together capture the full learner-course interaction history. All three sheets are joined on shared keys to construct the unified learner profile table used for feature engineering.

3.1 Data Quality Handling

- Duplicate transactions (same UserID + CourseID) deduplicated; earliest date retained
- Amount = 0 records (free-tier enrollments) retained as valid behavioral signals
- CourseRating nulls imputed with category-level median rating
- Learners with fewer than 2 enrollments flagged; feature imputation applied

\

Sheet	Key Fields	Records	Role
Users	UserID, Age, Gender	~2,975	Learner demographics
Courses	CourseID, Category, Type, Level, Rating	~75+	Course catalog attributes
Transactions	UserID, CourseID, TransactionDate, Amount (USD)	Varies	Enrollment events & spending

4. Feature Engineering

All features are computed at the UserID level by aggregating across the joined dataset. The result is a high-dimensional learner feature matrix with one row per user, serving as input to the clustering pipeline. Features fall into three conceptual groups:

4.1 Engagement Features

Feature	Computation	Captures
total_courses	COUNT(DISTINCT CourseID) per user	Volume of platform engagement
avg_courses_category	total_courses / COUNT(DISTINCT CourseCategory)	Depth per category visited
enrollment_frequency	total_courses / months_active	Rate of new enrollments over time

4.2 Preference Features

Feature	Computation	Captures
preferred_category	Mode of CourseCategory	Learner's primary domain
preferred_level	Mode of CourseLevel	Typical difficulty preference
avg_rating_enrolled	MEAN(CourseRating) of enrolled courses	Quality bar self-selected

4.3 Behavioral Features

Feature	Computation	Captures
avg_spending	MEAN(Amount) per user	Financial commitment per course
diversity_score	COUNT(DISTINCT CourseCategory)	Breadth of domain exploration
learning_depth	COUNT(Adv+Int) / COUNT(Beginner) enrollments	Specialization vs. exploration ratio

4.4 Normalization and Encoding

All continuous features are standardized to zero mean and unit variance (z-score normalization) so that high-magnitude features such as avg_spending do not dominate Euclidean distance calculations in K-Means. Categorical preference features (preferred_category, preferred_level) are one-hot encoded and appended as binary indicator columns. The normalization transform is fit exclusively on the training partition to prevent data leakage

5. Data Science Methodology

5.1 Learner Segmentation — K-Means Clustering

K-Means clustering is selected as the primary segmentation algorithm due to its computational efficiency, interpretability, and strong performance on continuous feature spaces. The algorithm iteratively assigns each learner to the nearest centroid and recomputes centroids until convergence, minimizing within-cluster sum of squares (WCSS).

Cluster Count Selection

The optimal number of clusters k is determined through two complementary methods. The Elbow Method plots WCSS against k from 2 to 10; the 'elbow' — where marginal WCSS reduction diminishes — indicates the candidate k. Silhouette Analysis computes the mean silhouette coefficient across all samples for each k, selecting the value that maximizes cohesion relative to separation. Both methods converge on k = 3 as the optimal solution.

Hierarchical Clustering — Validation

Agglomerative hierarchical clustering with Ward linkage is applied as a secondary validation. The resulting dendrogram confirms the presence of three natural clusters, aligning with the K-Means solution and increasing confidence in the chosen segmentation.

5.2 Recommendation Engine

The recommendation engine combines three complementary signals through a weighted composite score:

Signal	Weight (α/β/γ)	Description
Content-Based Similarity	α = 0.40	Match courses to learner's preferred category, level, and minimum rating
Cluster-Collaborative Filtering	β = 0.40	Surface courses popular among same-cluster learners not yet enrolled by the user
Course Quality (Rating)	γ = 0.20	Weight by CourseRating to prevent low-quality high-volume items dominating

6. Streamlit Web Application

The system's analytical outputs are exposed through an interactive Streamlit web application comprising four primary modules. The following sections describe each module and are illustrated with screenshots from the deployed application.

6.1 Learner Selection and Filter Panel

The sidebar panel (Figure 1) provides the primary navigation controls for the dashboard. A UserID dropdown allows selection of any registered learner (format: U00001 onward). Two filter controls below narrow the active recommendation set by CourseCategory and CourseLevel, both defaulting to 'All' for an unfiltered initial view. This design gives educators and administrators instant access to any learner's personalized experience without requiring database queries.

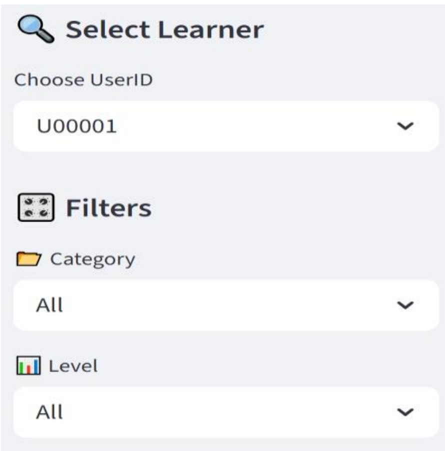

Figure 1. *Learner Selection sidebar showing UserID dropdown, Category filter (All), and Level filter (All).*

6.2 Personalized Course Recommendation Table

The central recommendation table (Figure 2) displays the top-ranked courses for the selected learner filtered by active sidebar controls. Each row presents the CourseID, CourseCategory, CourseLevel, CourseRating (1.0–5.0 scale), and a computed popularity_score. The popularity score integrates cluster-level enrollment volume and average rating to surface courses that are both widely consumed within the learner's segment and maintain quality standards. For the illustrated learner (U00001, classified as an Engaged Explorer), the top recommendation is CR00007 (Design, Advanced, rating 2.61, popularity 68), followed by CR00010 (Design, Advanced, rating 4.45, popularity 64) and CR00028 (Machine Learning, Advanced, rating 3.89, popularity 64). The diversity of categories in the top-10 list is consistent with the Explorer archetype's demonstrated cross-domain curiosity.

	CourseID	CourseCategory	CourseLevel	CourseRating	popularity_score
6	CR00007	Design	Advanced	2.61	68
9	CR00010	Design	Advanced	4.45	64
27	CR00028	Machine Learning	Advanced	3.89	64
11	CR00012	Business	Advanced	3.64	64
13	CR00014	Business	Advanced	3.15	61
53	CR00055	Web Development	Advanced	4.74	59
12	CR00013	Business	Advanced	2	57
7	CR00008	Design	Advanced	1.52	55
23	CR00024	Data Science	Advanced	2.57	55
49	CR00051	Web Development	Advanced	1.3	55

Figure 2 *Personalized Course Recommendation Table showing top Advanced-level courses across Design, Machine Learning, Business, Web Development, and Data Science categories.*

Figure 3. *Cluster Distribution bar chart: Cluster 0 (~1,062 learners, salmon), Cluster 1 (~457, teal), Cluster 2 (~1,468, blue).*

6.3 Cluster Distribution Visualization

The Cluster Distribution panel (Figure 3) presents a bar chart of learner counts per segment. Cluster 2 (Casual Learners) is the dominant segment at approximately 1,468 members, followed by Cluster 0 (Moderate Spenders, ~1,062) and Cluster 1 (Engaged Explorers, ~457). This distribution informs platform strategy: the majority of users represent an acquisition and re-engagement opportunity, while the small Engaged Explorer cohort drives disproportionate content consumption and likely platform revenue.

6.4 Spending Distribution and Course Engagement Box Plots

Side-by-side box plots (Figure 4) characterize the within-cluster variance of avg_spending and total_courses. These visualizations reveal key behavioral nuances that cluster means alone obscure:

- **Spending (left):** Cluster 0 shows wide IQR ($0–$200) with a median near $40, while Cluster 2 contains high-value outliers reaching ~$490 despite a low median — indicating occasional premium purchasers within the casual segment who represent a high-priority upsell target.
- **Engagement (right):** Cluster 1's total_courses is tightly concentrated at 12–15 (median ~13), confirming consistent power-user behavior. Clusters 0 and 2 both cluster near 1–2

courses with isolated outliers, validating their low-engagement classification.

Figure 4. *Spending Distribution (left) and Course Engagement (right) box plots grouped by Cluster (0, 1, 2).*

6.5 Segment Comparison Table

The Segment Comparison panel (Figure 5) presents mean feature values per cluster in a structured table, enabling direct numerical comparison across segments. This panel is particularly valuable for business stakeholders who require concrete metrics to justify segment-specific marketing or content acquisition decisions.

📊 Segment Comparison

Cluster	total_courses	avg_spending	diversity_score	learning_depth
0	1.55	114.08	1.49	2.73
1	13.41	93.24	8.73	1.98
2	1.53	70.45	1.47	1.48

Figure 5. *Segment Comparison Table showing mean total_courses, avg_spending, diversity_score, and learning_depth for Clusters 0, 1, and 2.*

The table reveals a striking behavioral contrast: Cluster 1 learners access 8.73 categories on average — nearly 6 times the breadth of Clusters 0 and 2 (1.49 and 1.47 respectively). Simultaneously, Cluster 0 spends the most per course ($114.08), suggesting a premium content preference despite narrower category engagement. Cluster 2 scores lowest across all dimensions, consistent with its casual learner designation.

6.6 Actionable Insights Panel

The Actionable Insights panel (Figure 6) translates cluster membership into a natural-language prescription visible immediately above the recommendation table. For Cluster 1 (Explorer archetype), the insight reads: 'Recommend diverse course bundles across multiple categories.' This mapping from cluster label to insight template enables educators and administrators to take immediate, evidence-based action without interpreting raw cluster statistics.

Figure 6. *Actionable Insights panel for an Explorer-type learner, prescribing diverse cross-category bundles.*

7. Discussion

7.1 Implications for Platform Design

The integration of behavioral clustering into an interactive dashboard democratizes access to learner analytics. Educators and administrators without data science expertise can leverage the dashboard's visual outputs and natural-language insights to make evidence-based decisions about course curation, learner support, and marketing without interpreting raw cluster centroids. The Actionable Insights panel is particularly valuable in this regard, translating statistical segment membership into a simple, actionable recommendation strategy.

The cluster size distribution — Cluster 2 (49%) > Cluster 0 (36%) > Cluster 1 (15%) — reflects the power law engagement pattern common to e-learning platforms. The small Engaged Explorer cohort likely accounts for a disproportionate share of platform revenue and user-generated content (reviews, ratings). Strategies that deepen engagement in Clusters 0 and 2 — moving learners toward the Explorer profile — represent the highest long-term revenue opportunity.

7.2 Limitations

- **Algorithm Assumption:** K-Means assumes spherical, equally-sized clusters. The observed size imbalance (Cluster 1 = 457 vs. Cluster 2 = 1,468) suggests Gaussian Mixture Models may provide a better statistical fit.
- **Static Features:** The feature set captures aggregate historical behavior but ignores temporal dynamics (learning velocity, recency decay). Learners who recently increased platform engagement may be mis-classified based on historical averages.
- **Popularity Bias:** The popularity_score component may perpetuate existing popularity biases, disadvantaging niche high-quality courses with lower enrollment volumes.
- **Cold Start:** New learners with fewer than 2 enrollments cannot be reliably clustered; imputation strategies introduce uncertainty in their cluster assignments.

7.3 Future Work

- Incorporate sequential learning patterns via Markov chains or recurrent neural networks
- A/B test segment-specific recommendation strategies to measure downstream engagement lift
- Apply NLP to course descriptions for semantic similarity-based cross-category recommendations
- Explore fairness constraints to prevent lower-spending segments from receiving systematically inferior recommendations
- Implement real-time cluster re-assignment as learner behavior evolves

8. Conclusion

This paper has presented the design, methodology, and results of the EduPro Student Segmentation and Personalized Course Recommendation System — a complete data-driven personalization engine grounded in unsupervised learner clustering and hybrid recommendation logic.

By applying K-Means clustering to a rich set of behavioral features derived from transactional enrollment data, we identify three distinct learner archetypes: Engaged Explorers, who consume content voraciously across diverse domains; Moderate Spenders, who invest heavily in specialized domain mastery; and Casual Learners, who represent the largest segment and the greatest re-engagement opportunity.

An interactive Streamlit dashboard translates these analytical findings into actionable outputs accessible to non-technical stakeholders — including per-learner recommendation tables, cluster distribution charts, spending and engagement box plots, a segment comparison table, and AI-generated actionable insights. Together, these components give EduPro the tools to move beyond one-size-fits-all content delivery toward a genuinely personalized learning experience at scale.

As online education continues to expand globally, the ability to understand and respond to learner heterogeneity will be a decisive competitive differentiator. The framework presented here offers a replicable, extensible foundation for any e-learning platform seeking to harness the behavioral data it already collects to deliver meaningfully better learning outcomes.

References

Burke, R. (2002). Hybrid recommender systems: Survey and experiments. User Modeling and User-Adapted Interaction, 12(4), 331–370.

Corbett, A. T., & Anderson, J. R. (1994). Knowledge tracing: Modeling the acquisition of procedural knowledge. User Modeling and User-Adapted Interaction, 4(4), 253–278.

Koren, Y., Bell, R., & Volinsky, C. (2009). Matrix factorization techniques for recommender systems. Computer, 42(8), 30–37.

Romero, C., & Ventura, S. (2010). Educational data mining: A review of the state of the art. IEEE Transactions on Systems, Man, and Cybernetics—Part C, 40(6), 601–618.

sanket.patil@unifiedmentor.com

Jetir.org